



Data Cleaning and Manipulation

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Introduction to Data Cleaning

Real-world data is messy

Data cleaning is the process of fixing incorrect, incomplete, and inconsistent data

It improves data quality before analysis

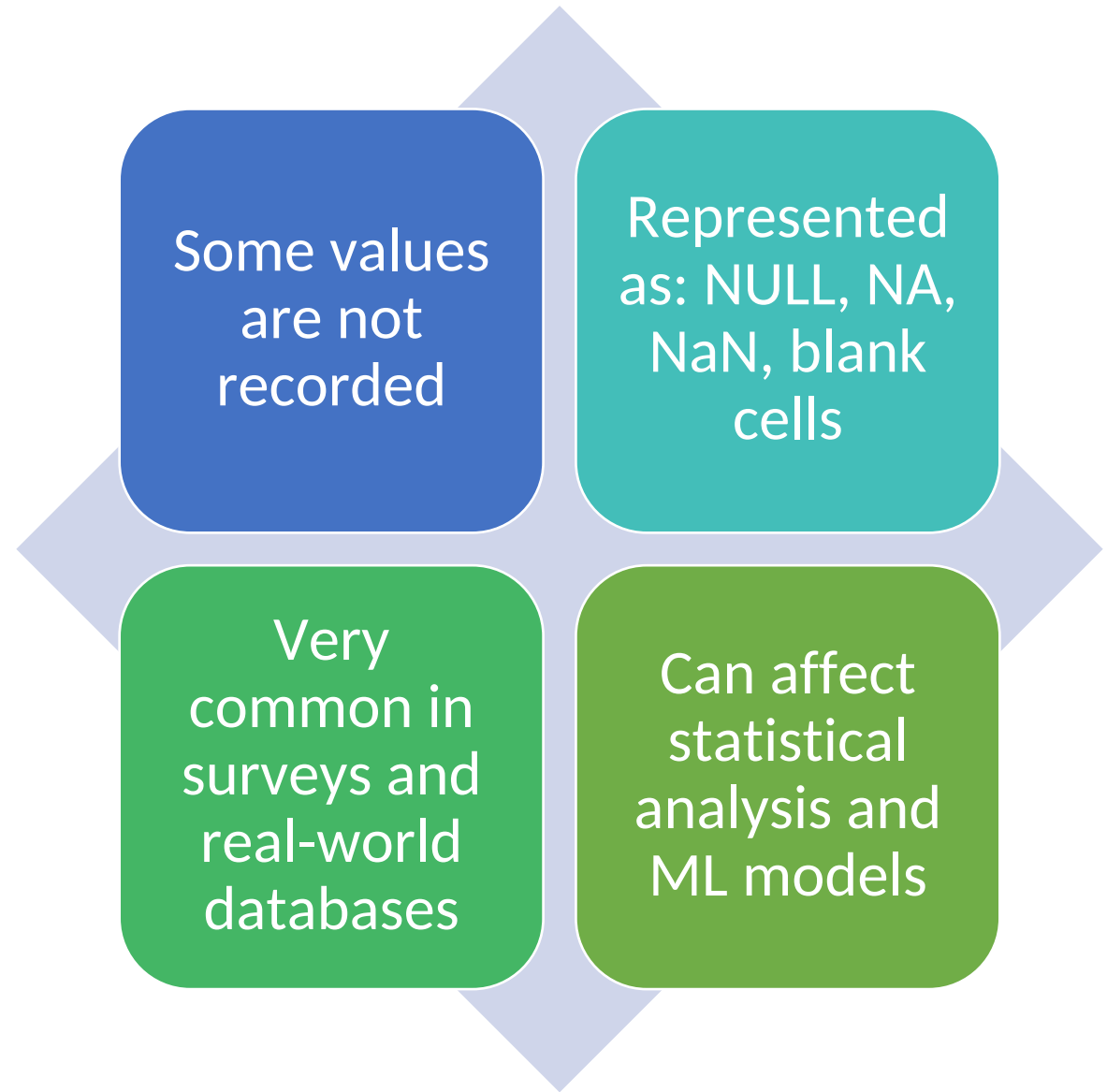
Also called Data Preprocessing

Essential before Machine Learning or Statistical Modeling

Common Data Quality Issues

- Missing values
- Duplicate records
- Outliers
- Incorrect data types
- Inconsistent formatting
- Noise and errors
- Invalid values

What is Missing Data?



A large orange circle on the left side of the slide, partially cut off by the edge.

Problems Caused by Missing Data

Reduces dataset size

Introduces bias

Decreases model accuracy

Can distort mean, variance and correlations

Handling Missing Data

Remove	Remove rows (Deletion)
Remove	Remove columns
Replace	Replace with mean / median / mode
Forward	Forward fill / backward fill
Predict	Predict missing values using ML

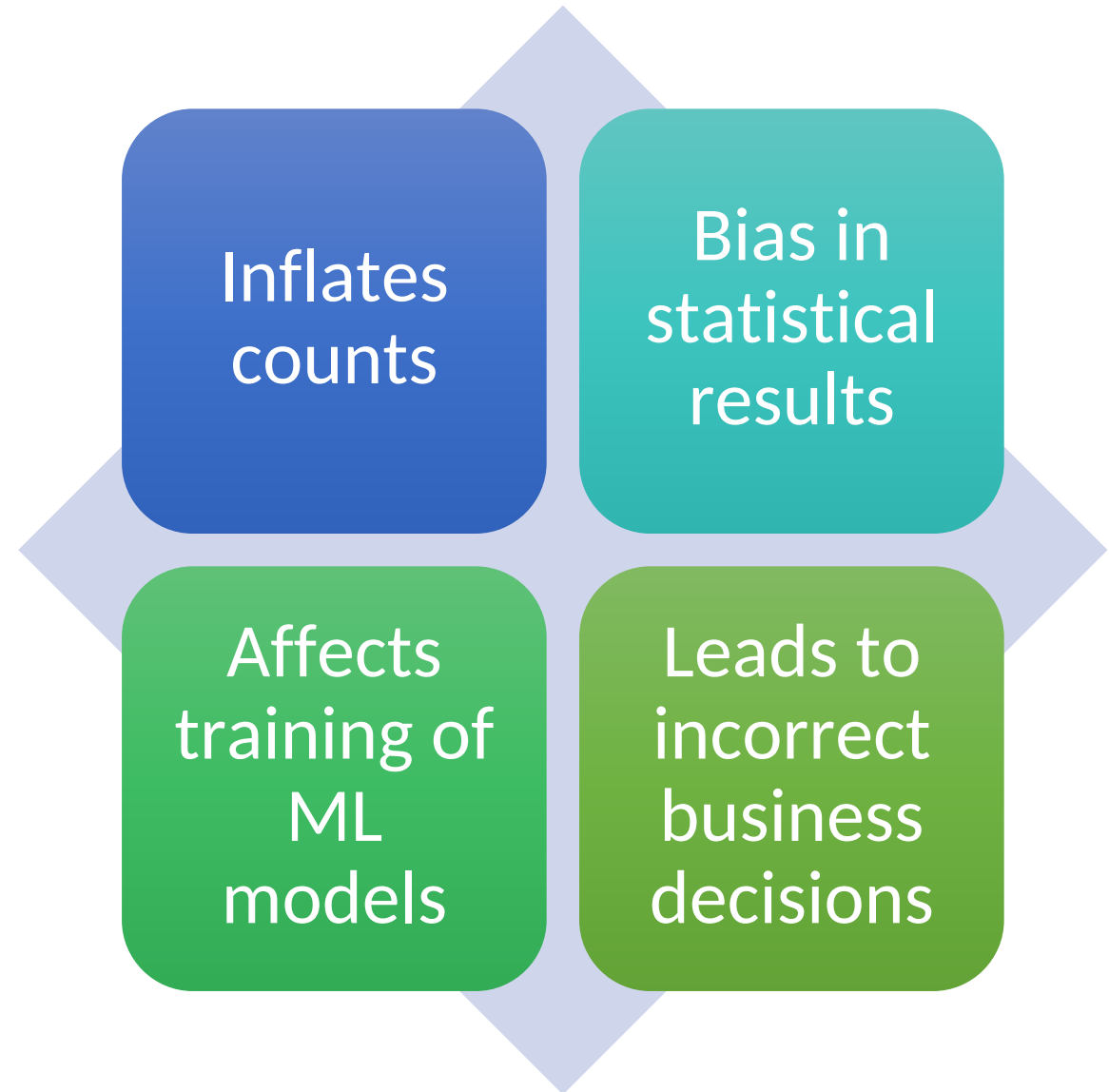
What are Duplicate Records?

Same record appears multiple times

Can happen during data entry or merging datasets

Creates false representation of frequency

Problems Caused by Duplicates



Handling Duplicates

1

Identify using
unique keys

2

Remove exact
duplicates

3

Check partial
duplicates

4

Keep first
occurrence or
latest record

What are Outliers?

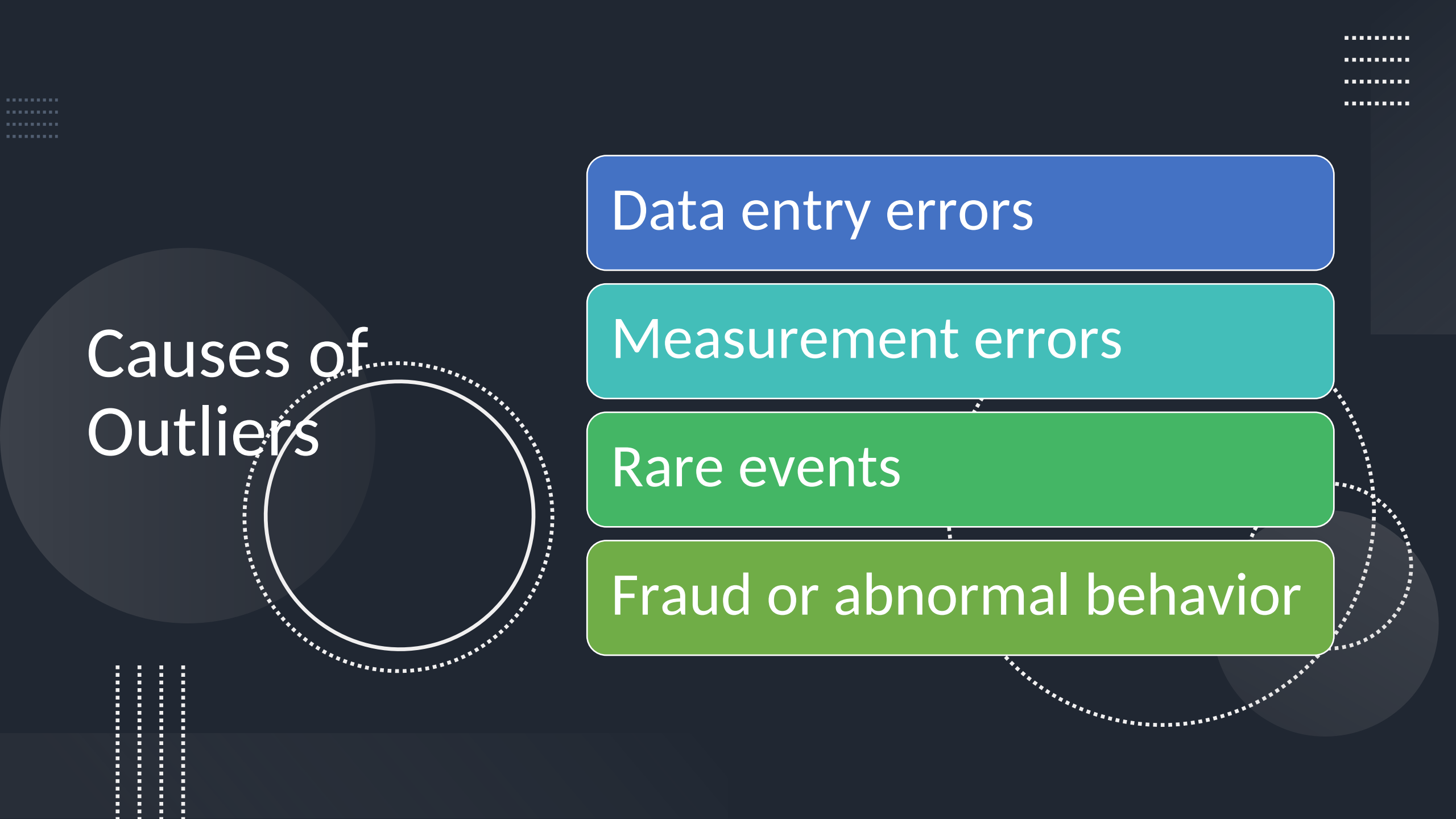
Extreme values that are very different

Example: Age = 200

Can be real or error

Can distort statistical analysis

Causes of Outliers



Data entry errors

Measurement errors

Rare events

Fraud or abnormal behavior

Detecting Outliers



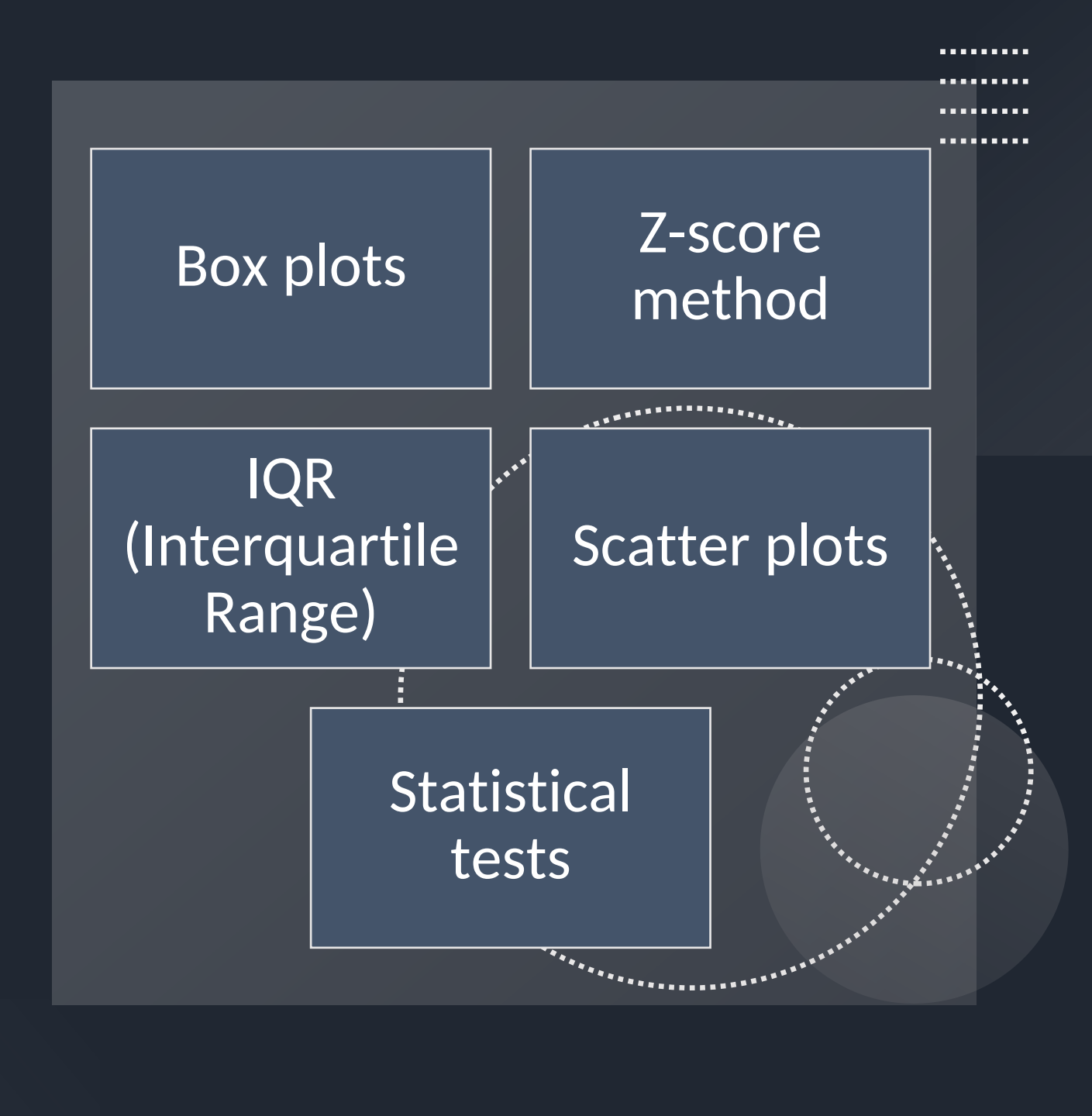
Box plots

Z-score
method

IQR
(Interquartile
Range)

Scatter plots

Statistical
tests



Handling Outliers

Remove

Remove if error

Cap

Cap values (Winsorization)

Transform

Transform data (log transformation)

Keep

Keep if meaningful



Other Data Cleaning Issues

Inconsistent Data Formats

Date formats (DD/MM/YYYY vs MM/DD/YYYY)

Uppercase vs lowercase

Different category labels

Extra spaces

Incorrect Data Types

Numbers stored as text

Date stored as string

Boolean stored as 0/1 or Yes/No

Must convert to correct type

Noise and Invalid Data

Typing mistakes

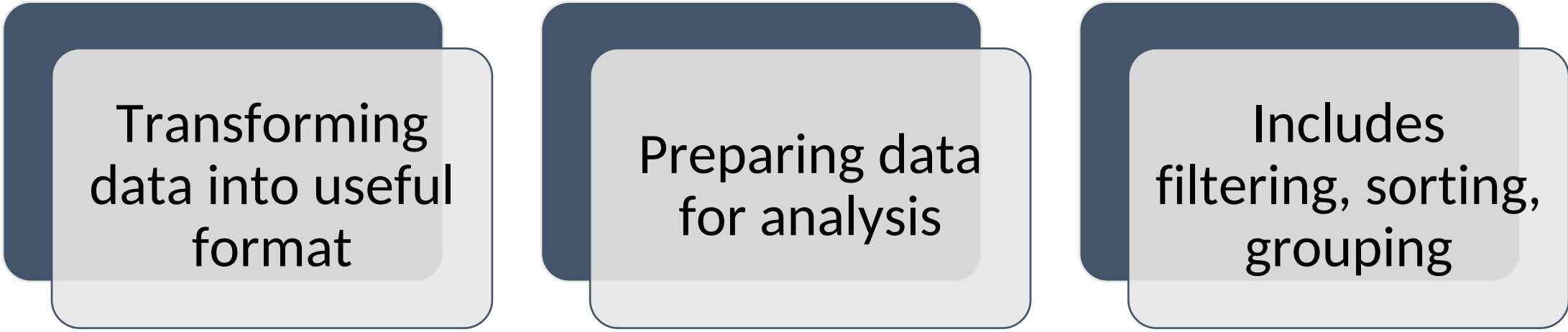
Impossible values (Negative age)

Broken sensor data

Use validation rules



What is Data Manipulation?



Transforming
data into useful
format

The diagram consists of three identical rounded rectangular boxes arranged horizontally. Each box has a dark blue outer layer and a light gray inner layer. The text is centered within the light gray inner layer. The first box contains the text 'Transforming data into useful format', the second box contains 'Preparing data for analysis', and the third box contains 'Includes filtering, sorting, grouping'. A solid yellow horizontal line is located at the bottom of the slide.

Preparing data
for analysis

Includes
filtering, sorting,
grouping

Common Data Manipulation Operations

Filtering	Filtering rows
Selecting	Selecting columns
Sorting	Sorting data
Grouping	Grouping and aggregation
Creating	Creating new variables
Merging	Merging datasets

Data Cleaning + Manipulation in Practice



Real projects
spend 70–80%
time cleaning data



Cleaning improves
model
performance



Manipulation
prepares data for
insights



Data quality =
Better decisions



Foundation of Data
Science